

Transient hepatic reconstitution of trophic factors enhances aged immunity

<https://doi.org/10.1038/s41586-025-09873-4>

Received: 4 October 2024

Accepted: 5 November 2025

Published online: 17 December 2025

Open access

 Check for updates

Mirco J. Friedrich^{1,2,3,4,5}, Julie Pham^{1,2,3,4,5}, Jiakun Tian^{2,6}, Hongyu Chen^{2,6}, Jiahao Huang^{2,6}, Niklas Kehl², Sophia Liu^{2,7}, Blake Lash^{1,2,3,4,5}, Fei Chen^{2,8,9}, Xiao Wang^{2,6,10}, Rhiannon K. Macrae^{1,2,3,4,5} & Feng Zhang^{1,2,3,4,5}✉

Ageing erodes human immunity, in part by reshaping the T cell repertoire, leading to increased vulnerability to infection, malignancy and vaccine failure^{1–3}. Attempts to rejuvenate immune function have yielded only modest results and are limited by toxicity or lack of clinical feasibility^{1,3–5}. Here we show that the liver can be transiently repurposed to restore age-diminished immune cues and improve T cell function in aged mice. These immune cues were found by performing multi-omic mapping across central and peripheral niches in young and aged animals, leading to the identification of Notch and Fms-like tyrosine kinase 3 ligand (FLT3L) pathways, together with interleukin-7 (IL-7) signalling, as declining with age. Delivery of mRNAs encoding Delta-like ligand 1 (DLL1), FLT3L and IL-7 to hepatocytes expanded common lymphoid progenitors, boosted de novo thymopoiesis without affecting haematopoietic stem cell (HSC) composition, and replenished T cells while enhancing dendritic cell abundance and function. Treatment with these mRNAs improved peptide vaccine responses and restored antitumour immunity in aged mice by increasing tumour-specific CD8⁺ infiltration and clonal diversity and synergizing with immune checkpoint blockade. These effects were reversible after dosing ceased and did not breach self-tolerance, in contrast to the inflammatory and autoimmune liabilities of recombinant cytokine treatments^{6,7}. These findings underscore the promise of mRNA-based strategies for systemic immune modulation and highlight the potential of interventions aimed at preserving immune resilience in ageing populations.

Ageing has a profound effect on the immune system, including the T cell repertoire, leading to reduced immune resilience^{1–3}. Central to this decline in humans and most other mammals is the involution of the thymus. Thymic involution curtails naive T cell output, contracts T cell receptor (TCR) repertoire diversity and blunts primary responses, whereas peripheral T cells accrue dysfunctional states that heighten susceptibility to infection, vaccine failure and cancer.

Efforts to counter immune ageing have primarily focused on reversing thymic involution through hormones⁸, cytokines⁹, small molecules¹⁰ and heterochronic parabiosis¹¹, or by directly modulating haematopoiesis¹². Although these strategies have provided valuable insights into immune ageing, they have been limited by effect size, toxicity or clinical feasibility^{4,5}.

Here we describe an approach for reconstituting thymus-derived factors in the liver to address age-related immune decline (Fig. 1a). We first identified signalling pathways in the thymus and peripheral blood T cells that decline with age. We then delivered mRNAs encoding these factors (DLL1, FLT3-L and IL-7) to the liver using lipid nanoparticles (LNPs). We found that this approach significantly improved immune response in ageing mice in both vaccination and cancer immunotherapy

models with no adverse side effects or evidence of increased autoimmunity. These results highlight the potential of this approach to improve immune function and, more broadly, to use the liver as a transient ‘factory’ for replenishing factors that decline with age.

T cell support signals decline with age

A functional T cell repertoire depends on pro-survival and trophic cues: cytokines, hormones and self-peptide MHC, which are abundant in healthy individuals but diminish with age, contributing to immune decline^{13–15}.

To nominate potentially immune restorative cues, we profiled trophic signalling across the mouse lifespan using single-cell RNA sequencing (scRNA-seq) or TCR-seq of peripheral blood T cells and spatially resolved Slide-TCR-seq on the thymus, the principal organ of T cell production¹⁶. After quality control, the dataset comprised approximately 97,000 circulating T cell transcriptomes (CD4⁺ and CD8⁺) and 47 spatial arrays covering approximately 1.26 million thymic positions spanning the day of birth to 90 weeks of age (Extended Data Fig. 1a–h). Ageing shifted T cell states, with loss of naive (*Tcf7*, *Sell* and *Ccr7*) and

¹Howard Hughes Medical Institute, Cambridge, MA, USA. ²Broad Institute of MIT and Harvard, Cambridge, MA, USA. ³McGovern Institute for Brain Research at MIT, Cambridge, MA, USA.

⁴Department of Brain and Cognitive Science, Massachusetts Institute of Technology, Cambridge, MA, USA. ⁵Department of Biological Engineering, Massachusetts Institute of Technology, Cambridge, MA, USA. ⁶Department of Chemistry, Massachusetts Institute of Technology, Cambridge, MA, USA. ⁷Ragon Institute of MGH, MIT and Harvard, Cambridge, MA, USA. ⁸Harvard Stem Cell Institute, Cambridge, MA, USA. ⁹Department of Stem Cell and Regenerative Biology, Harvard University, Cambridge, MA, USA. ¹⁰Stanley Center for Psychiatric Research, Broad Institute of MIT and Harvard, Cambridge, MA, USA. ✉e-mail: zhang@broadinstitute.org

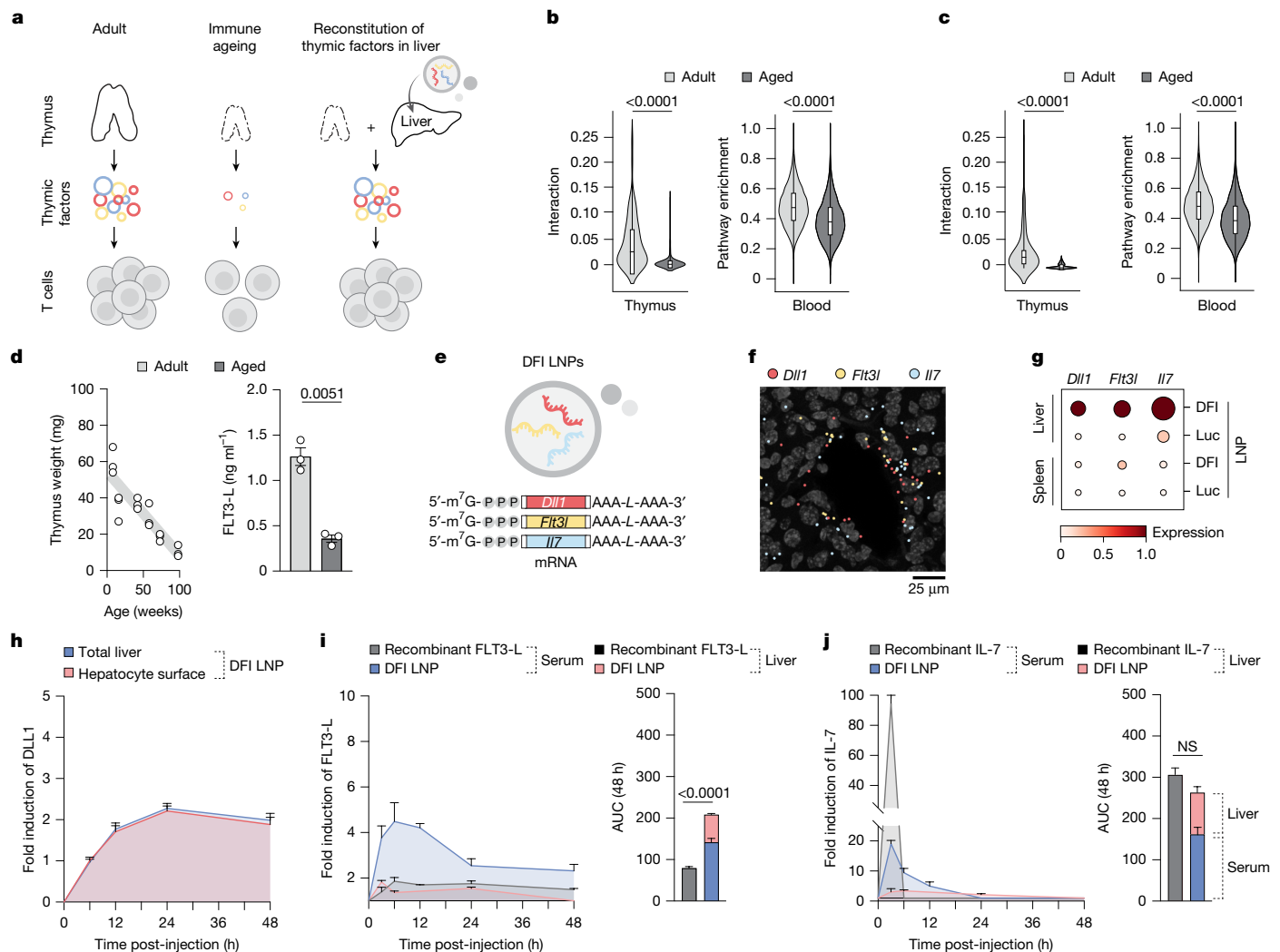


Fig. 1 | Hepatic reconstitution of declining T cell signalling factors to restore immune signalling in ageing. **a**, Overview of the approach to restore age-declining immune trophic cues by hepatic expression of *Dll1*, *Flt3l* and *Il7* mRNAs. **b**, Spatial ligand–receptor interactions between thymic cortical epithelial cells (cTECs) and thymocytes decline with age (left), and ssGSEA shows reduced Notch pathway activity in circulating T cells (right). $n = 47$ spatial arrays and 96,683 blood T cell transcriptomes across 21 ages. Data are represented as violin plots with median + interquartile range. Statistical significance was determined by Mann–Whitney tests. **c**, cTEC–T cell IL-7 interaction (left) and downstream pathway in circulating T cell (right) activities are likewise diminished with age. $n = 47$ spatial arrays and 96,683 blood T cell transcriptomes across 21 ages. Data are represented as violin plots with median + interquartile range. Statistical significance was determined by Mann–Whitney tests. **d**, Thymus weight decreases with age ($n = 18$; 3 per timepoint). Interstitial FLT3-L levels are reduced in aged thymus by ELISA ($n = 3$ per group). Data are mean $\pm$ s.e.m.; statistical significance was determined by a two-tailed unpaired Student's *t*-test. **e**, mRNA (DFI; *Dll1*, *Flt3l* and *Il7*) constructs formulated in

SM-102 LNPs. **f**, Representative RIBOmap images 6 h post-DFI show robust ribosome-bound transcripts in the liver. A representative image from three imaged DFI-treated animals is shown. **g**, Single-cell quantification: translating *Dll1*, *Flt3l* and *Il7* in the liver and spleen by RIBOmap ($n = 1$ for Luc and $n = 3$ for DFI). **h**, Immunofluorescence of DLL1 protein over 0–48 h after 5 μ g DFI reveals transient induction in total liver and hepatocyte surface (phalloidin co-stain). Fold induction from baseline (0 h) is shown. $n = 32$ fields of view from $n = 3$ animals per time point per condition. Data are mean $\pm$ s.e.m. **i**, ELISA for FLT3-L levels in serum and the liver after 10 μ g recombinant FLT3-L or 5 μ g DFI at 3–48 h. Liver concentrations were normalized to liver weight; fold change from 0 h is shown. $n = 3$ animals per time point per compartment per condition. Data are mean $\pm$ s.e.m.; area under the curve (AUC) over 48 h compared by a two-tailed unpaired Student's *t*-test. **j**, ELISA for IL-7 in serum and the liver after 10 μ g recombinant IL-7 or 5 μ g DFI at 3–48 h with liver normalization and fold change from 0 h. $n = 3$ animals per time point per compartment per condition. Data are mean $\pm$ s.e.m.; 48-h AUC compared by a two-tailed unpaired Student's *t*-test. NS, not significant.

stem-like populations (*Bcl11b*, *Lef1*, *Id3* and *Sox4*), and expansion of (virtual) memory (*Cd44*, *Eomes*, *Gzmb* and *Tbx21*) and exhaustion-like phenotypes (*Pdcd1*, *Havcr2*, *Ctla4*, *Lag3*, *Tigit*, *Gzmk* and *Entpd1*), consistent with previous mouse and human studies^{17,18} (Extended Data Figs. 1e,h,i and 2).

Using spatial proximity to infer bona fide cell–cell interactions (rather than co-regulation alone), we used permutation-based null models to identify age-dependent receptor–ligand pairs between thymocytes and thymic epithelial cells (TECs). Cortical TEC–T cell signalling declined markedly with age, whereas medullary TEC–T cell

interactions were relatively preserved (Extended Data Fig. 3a,b). Integrating spatially informed interaction analysis with single-sample Gene Set Enrichment Analysis (ssGSEA) of peripheral blood-circulating T cells revealed age-linked attenuation of Notch1/3 and IL-7 signalling (Fig. 1b,c and Extended Data Fig. 3c,d). Although thymic stromal cells primarily act locally to support T cell maturation and selection, several of their soluble products accumulate in the thymus, enter the circulation and contribute to systemic immune homeostasis^{13,19}. Consistently, intrathymic as well as circulating T cells from aged mice showed reduced expression of Notch, IL-7 and FLT3-L target genes and diminished

downstream activity (Fig. 1b,c and Extended Data Fig. 3e,f). Soluble FLT3-L, produced in part by intrathymic fibroblasts and required to sustain T cell function in aged and post-transplant settings, also declined in thymic homogenates^{20,21} (Fig. 1d and Extended Data Fig. 3g).

These findings motivated us to transiently reconstitute age-diminished immune signalling using three immune trophic factors – DLL1 (to activate Notch), FLT3-L and IL-7 – in aged hosts. We selected DLL1 over DLL4 based on a more favourable safety and immunological profile: DLL4 is linked to angiogenesis and vascular remodelling and can overly restrict lymphoid differentiation, whereas we hypothesized that DLL1 supports T cell development without suppressing B cell output^{22,23}.

Reconstituting immune cues in the liver

To bypass the structural and functional constraints of the involuted thymus, we reconstituted the identified signalling pathways ectopically in the liver. The unique haemodynamics and anatomical features of the liver enable priming and maintenance of adaptive immunity, and its protein-synthesis capacity is preserved even at advanced ages, making it a suitable site to modulate circulating T cells^{24–26}.

We chose mRNA delivery over recombinant proteins because recombinant cytokines clear rapidly, necessitating frequent high-dose administrations that often result in significant toxicity^{6,7}. By contrast, mRNA allows for more controlled, transient protein production, and recent advances have further optimized mRNA stability and functionality while minimizing immune-related side effects for in vivo applications^{6,27–29}. In addition, canonical Notch ligands are transmembrane proteins and require cell–cell contact, precluding soluble delivery. We therefore packaged mRNAs encoding DLL1 (*Dll1*), FLT3-L (*Flt3l*) and IL-7 (*Il7*) (collectively, DFI), or firefly luciferase (Luc) or GFP controls, in an SM-102 LNP formulation (Fig. 1e, Extended Data Fig. 4a–e and Supplementary Fig. 1a–e). All mRNAs were m¹Ψ modified and 5′-m⁷GpppNm capped. Primary hepatocytes expressed DLL1 on the surface and secreted biologically active IL-7 and FLT3-L into the supernatant after transfection (Extended Data Fig. 4a–c). Systemic LNP administration predominantly targeted the liver with minimal translation in other tissues (Fig. 1f,g, Extended Data Fig. 4d,e and Supplementary Fig. 1f,g). In situ profiling of ribosome-bound (rather than endosome-trapped) transcripts (RIBOmap) confirmed robust translation of all three DFI mRNAs in hepatocytes in vivo (Fig. 1f and Extended Data Fig. 4d,e), indicating hepatic expression and release with negligible off-target translation³⁰.

We next assessed pharmacokinetics and safety. *Dll1* mRNA induced sustained DLL1 on the surface of hepatocytes for approximately 48 h (ref. 31) (Fig. 1h and Extended Data Fig. 4f). mRNA-encoded FLT3-L achieved comparable serum half-life but higher single-dose systemic levels than recombinant FLT3-L (Fig. 1i). Intravenous recombinant IL-7 produced a sharp serum spike, whereas mRNA-encoded IL-7 yielded lower-amplitude, sustained levels with an approximately tenfold lower peak over 24 h (Fig. 1j). Because secreted IL-7 binds to heparan-sulfate proteoglycans, a substantial fraction of mRNA-delivered IL-7 remained in the hepatic extracellular matrix up to 24 h after serum levels normalized (Fig. 1j), potentially providing a compartmentalized source of IL-7 for cells traversing the sinusoids, where they also encounter hepatocyte-bound DLL1 (refs. 32,33).

Despite prolonged bioavailability of the three factors, we observed no changes in body weight, transaminases or liver function, and only minimal hepatic inflammation on histopathology after 4 weeks of DFI mRNA–LNPs in aged mice (Extended Data Fig. 4g–i), consistent with previous studies of SM-102 formulations^{6,34}. By contrast, recombinant FLT3-L and IL-7 administered on the same schedule (two doses per week for 28 days) induced marked elevations of GM-CSF, IL-10, IL-12 and IL-1β, underscoring the greater risk of systemic inflammation with recombinant cytokines (Supplementary Tables 2 and 3 and Supplementary Fig. 1h,i).

DFI mitigates immune ageing phenotypes

Ageing is accompanied by a loss of naive T cells and accumulation of (virtual) memory and exhaustion-like states, most pronounced in CD8⁺ T cells and driven by thymic involution, chronic antigen exposure and inflammation^{1,17,18}. We observed these shifts in our longitudinally profiled cohorts (Fig. 2a and Extended Data Figs. 1i and 2b,c). This bias towards the memory phenotype in ageing compromises responses to new antigens. We therefore asked whether DFI rebalances naive T cell representation.

Aged mice (72 weeks old) received DFI mRNA–LNPs or Luc mRNA–LNPs twice weekly for 28 days. DFI (but not individual factors) increased both frequency and absolute number of circulating naive (CD44⁺CD62L⁺) CD4⁺ and CD8⁺ T cells (Fig. 2b and Extended Data Fig. 5). Repeated administration of recombinant IL-7 has been shown to promote the proliferation of mature T cell subsets^{35–37}. In line with this, *Il7* mRNA alone expanded effector-memory (CD44⁺CD62L⁺) cells; however, full DFI did not increase memory subsets (Fig. 2c and Extended Data Fig. 5c–e). Consequently, the naive-to-memory ratio rose in the blood and spleen of DFI-treated animals, indicating complementary or moderating activity of the three signals when delivered in combination (Fig. 2d and Extended Data Fig. 5b).

To delineate the origin of these DFI-induced shifts in the T cell compartment of aged mice, we next analysed the spleen, peripheral blood, bone marrow and thymus of adult and aged mice. Deep bulk V(D)J sequencing showed no increase in clonality following DFI conditioning (Fig. 2e), arguing against homeostatic proliferation and favouring enhanced thymic output as a driver of the observed increase in naive T cell counts. Although aged thymuses displayed reduced mass and cellularity relative to young controls, 28 days of DFI partially restored both (Extended Data Fig. 6a–c). Although overall distributions of double-negative (DN), double-positive or single-positive thymocytes were preserved, DFI selectively expanded early DN1–DN3 thymocytes – the stages of early T lineage commitment – and induced *Rag2* in thymocytes of *Rag2*–eGFP mice within 12 h, followed by increased mature single-positive CD4⁺ and CD8⁺ thymocytes towards youthful levels after 28 days of treatment (Fig. 2f and Extended Data Figs. 6d–f and 7a–c). Consistent with these data, TCR excision circles in peripheral blood were elevated (Fig. 2g), and *Nur77* expression was elevated in circulating T cells (Extended Data Fig. 7d–f), indicative of an increase in recent thymic emigrants.

Because thymic output depends on progenitor supply, we examined haematopoiesis. With ageing, HSCs expand in number but increasingly adopt a myeloid-biased differentiation program^{12,38,39}; we recapitulated HSC expansion and CD150^{high} myeloid-biased HSC enrichment with age, which DFI did not reverse (Extended Data Fig. 8a–d). Multipotent progenitors were largely unchanged aside from an increase in lymphoid-primed multipotent progenitors after DFI (Extended Data Fig. 8b). By contrast, DFI robustly expanded common lymphoid progenitors (CLPs), which are markedly depleted with age, in bone marrow and increased the number of circulating CCR9⁺ (but not CCR7⁺) CLPs, which are associated with preferential thymus homing^{38,40} (Extended Data Figs. 8e–i and 9a–c). CCR9 surface levels on CLPs rose from the bone marrow to the thymus in DFI-treated mice, mirroring the dynamics found in young animals and absent in aged controls (Extended Data Fig. 9c–e), consistent with a peripheral priming of circulating CLPs that potentially augments thymic entry. Together, these data support a model in which liver-expressed DLL1, together with systemic FLT3-L and IL-7, boosts CLP production, survival and recruitment to the thymus, thereby increasing thymopoiesis. Thus, DFI enhances naive T cell output by amplifying committed lymphoid progenitors and facilitating intrathymic maturation, without reprogramming HSC composition.

Beyond T cells, DFI mitigated ageing-associated phenotypes in antigen-presenting and B cell compartments. DFI preferentially expanded splenic conventional type 1 dendritic cells (cDC1s), which

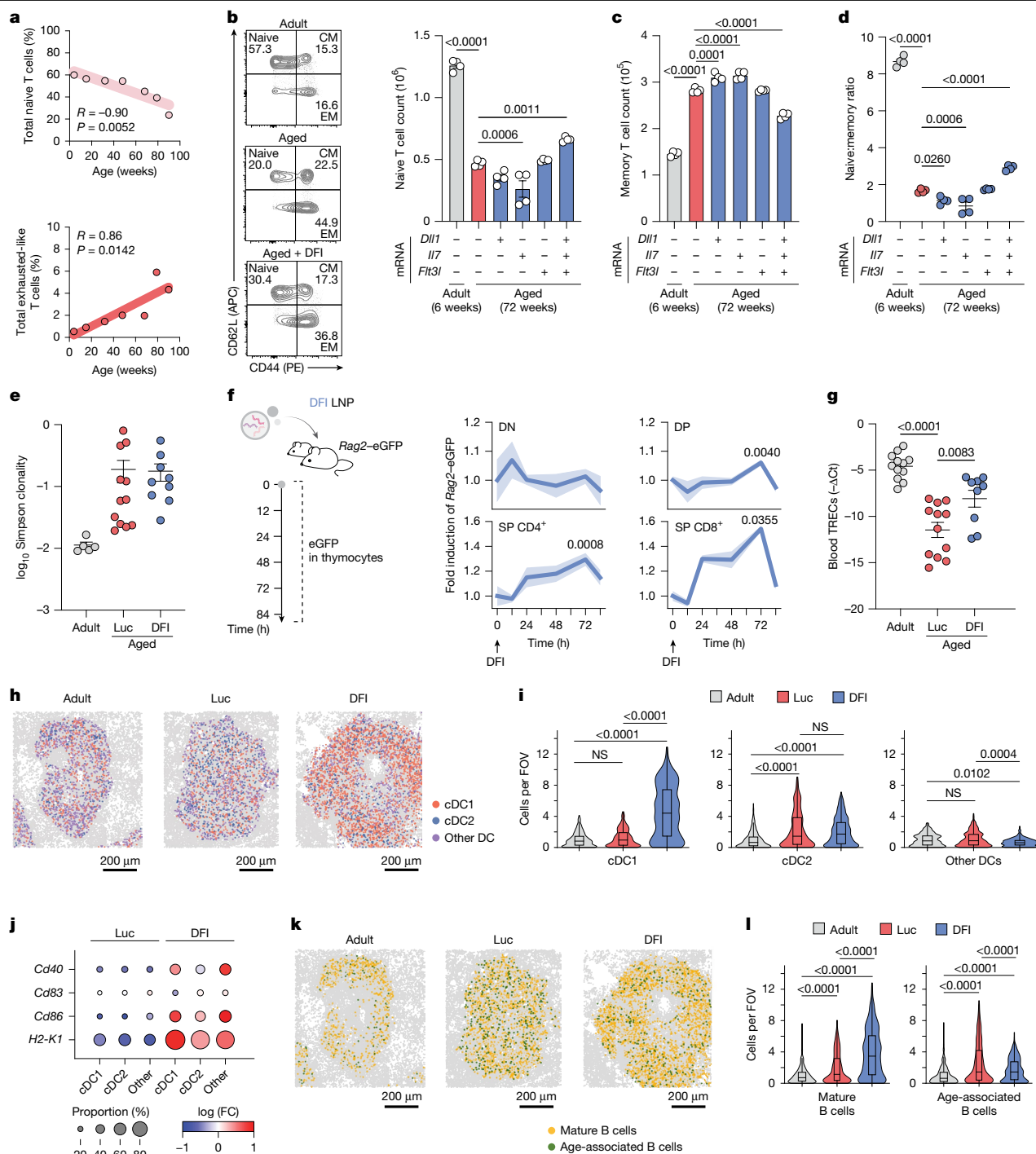


Fig. 2 | DFI treatment mitigates ageing-associated immune features. **a**, Age-related decline in circulating naive T cells (*Tcf7*, *Sell* and *Ccr7*) and accumulation of exhausted-like T cells (*Pdcd1*, *Ctla4*, *Lag3*, *Tiglt*, *Entpd1* and *Tnfrsf9*) in mouse peripheral blood across age. Linear regression (red lines) shows Pearson's R and significance versus age. **b–d**, Flow cytometry of spleens from adult (6 weeks) and aged (72 weeks) mice treated twice weekly with DFI or control (Luc) mRNA-LNPs for 28 days. Absolute numbers of naive T cells (CD44^{low}CD62L^{high}; **b**) and memory T cells (CD44^{high}CD62L^{low}; **c**), and naive-to-memory T cell ratio (**d**) are shown. $n = 4$ per group. Data are mean $\pm$ s.e.m.; one-way ANOVA with Dunnett's post-hoc test. CM, central memory; EM, effector memory. **e**, TCR repertoire diversity from bulk V(D)J sequencing ($n = 5$ adult, 12 aged + Luc and 9 aged + DFI). Data are mean $\pm$ s.e.m. **f**, Rag2-eGFP tracing of thymocyte maturation (DN, double-negative; DP, double-positive; SP CD4⁺ and SP CD8⁺) demonstrating increased thymopoiesis after DFI treatment. $n = 3$ mice per time point. Negative control refers to Rag2-eGFP^{-/-}. Data are mean $\pm$ s.e.m. (error bands); repeated-measures two-way ANOVA with Dunnett's post-hoc test was used.

g, Quantification of TCR excision circles (TRECs) in peripheral blood indicating recent thymic emigrants ($n = 12$ adult, 12 aged + Luc and 9 aged + DFI); data are mean $\pm$ s.e.m.; one-way ANOVA with Tukey post-hoc test. **h**, Representative spatial STARmap projections of splenic dendritic cell (DC) subtypes from adult, Luc-treated and DFI-treated mice. **i**, Quantification of cDC1, cDC2 and other DC subsets per field of view (FOV; adult = 404, Luc = 386 and DFI = 408 FOVs from one animal per condition); data are represented as violin plots with median + interquartile range; one-way ANOVA with Tukey post-hoc test. **j**, Differential expression of activation markers (*Cd40*, *Cd83*, *Cd86* and *H2-K1*) in the indicated DC subsets. FC, fold change. **k**, Spatial STARmap projections of splenic B cell subtypes. **l**, Relative abundance of mature (*Cd19*⁺, *Ms4a1*⁺, *Cd22*⁺ and *Cd40*⁺) and age-associated (*Cd19*⁺, *Ms4a1*⁺, *Itgax*⁺ and *Tbx21*⁺) B cells per FOV (adult = 404, Luc = 386 and DFI = 408 FOVs from one animal per condition); data are represented as violin plots with median + interquartile range; one-way ANOVA with Tukey post-hoc test.

are critical for antigen cross-presentation and decline in number and co-stimulatory function with age, probably by sustained FLT3-L-dependent differentiation from FLT3⁺ progenitors^{20,41,42} (Fig. 2h,i and Supplementary Fig. 2a–c). In situ-seq of spleens by STARmap revealed cDC1 enrichment within periaarteriolar lymphoid sheaths and a concomitant upregulation of *H2-K1* and co-stimulatory molecules (CD40, CD83 and CD86), consistent with improved priming capacity⁴³ (Fig. 2j and Supplementary Fig. 2a–c). In addition, we observed a reduced abundance of splenic age-associated B cells (CD19⁺, *Ms4a1*⁺, *Ilgax*⁺ and *Tbx21*⁺) and an expansion of mature follicular-like B cells (CD19⁺, *Ms4a1*⁺, *Cd22*⁺ and *CD40*⁺) localized to B cell follicles following DFI treatment^{44,45} (Fig. 2k,l and Supplementary Fig. 2a–e).

DFI strengthens vaccine responses

On the basis of the above findings, we next tested whether DFI could restore adaptive immune function in aged animals. As a test case, we looked at vaccine response, a T cell-mediated process known to be blunted by ageing. In an adjuvanted ovalbumin (OVA) prime–boost model in adult (6 weeks) and aged (72 weeks) mice, aged cohorts showed fewer OVA-specific CD8⁺ T cells in the spleen and blood upon vaccination, impaired antigen-driven proliferation and reduced IL-2 and IFN γ production on recall, approaching levels of sham-vaccinated mice (Supplementary Fig. 3a–d). These deficiencies align with the impaired T cell-mediated vaccine responses commonly observed in older animals and humans^{1,46}. Pre-conditioning aged mice with individual factors or the full DFI combination before vaccination showed that *Il7* mRNA alone increased total T cell counts, whereas only DFI increased both total T cells and the frequency of OVA-specific CD8⁺ cells, yielding approximately twofold more antigen-specific CD8⁺ T cells in the spleen and a similar increase in blood^{35–37} (Fig. 3a–d and Supplementary Fig. 3e,f).

In addition to the reduced generation of new immune cells, ageing is characterized by the progressive shift to dysfunctional states by existing cells^{2,17}. Aged mice were shown to accumulate PD1^{high}CD62L[−] T cells, which we confirmed using scRNA-seq and flow cytometry in our longitudinal cohorts^{12,47} (Fig. 2a and Extended Data Fig. 1e,i). This phenotype expanded further with repeated vaccination, unlike in adults (Fig. 3f). DFI conditioning preserved a higher naive fraction post-vaccination and reduced PD1^{high}CD62L[−] cells (with a reduced effect from IL-7 alone), together producing a more balanced T cell composition (Fig. 3e–g and Supplementary Fig. 3g). Functionally, CD8⁺ T cells from DFI-treated mice generated higher levels of IL-2 and IFN γ upon antigen-specific restimulation, whereas OVA-specific CD4⁺ frequencies and cytokines were largely unchanged (Fig. 3h and Supplementary Fig. 4a–e), suggesting that DFI preferentially supports CD8⁺ T cell responses.

In a longitudinal vaccination study across the lifespan of C57BL/6J mice, we found that DFI treatment increased vaccine-induced T cell counts in aged mice to levels comparable with those seen in much younger mice, effectively rejuvenating their response by approximately 24 weeks (Fig. 3i). Together, the full DFI combination, but not its single components, counteracts age-related defects in CD8⁺ vaccine responses, consistent with the broader principle that aged immunity retains the capacity to mount robust immune responses when sufficiently stimulated⁴⁸.

DFI rejuvenates antitumour responses

We next tested whether DFI improved responsiveness to tumour immunotherapy, which also diminishes with ageing. In B16-OVA melanoma and MC38-OVA colon carcinoma, we found that aged mice showed faster tumour progression and poorer survival than adults; immune checkpoint inhibition (ICI) with PDL1 blockade that controlled tumours in adults conferred little benefit in aged cohorts (Supplementary Fig. 5a–h), mirroring reports of age-dependent efficacy of PD1 or CTLA4 ICI^{48–50}.

Pre-conditioning aged hosts with DFI for 28 days followed by a 72-h washout to control for direct antitumour effects improved endogenous control of MC38-OVA, increasing spontaneous rejection rates and prolonging survival (Supplementary Fig. 5e–h). In the more aggressive B16-OVA model, DFI drove complete rejection in 40% of aged mice, whereas all controls succumbed within 3 weeks despite anti-PDL1 treatment (Fig. 4a–d). Co-administration at treatment onset similarly delayed progression and improved survival over PDL1 blockade alone in aged animals with established tumours (Fig. 4e–h).

In young mice, ICI confers antitumour immunity primarily through expansion and activation of tumour-specific CD8⁺ T cells. Ageing impairs this process, and elderly mice and humans show diminished CD8⁺ T cell infiltration and intratumoural effector function. To probe the underlying mechanism of the observed antitumour effects, we analysed tumour-infiltrating lymphocytes (TILs) on day 12 post-implantation, before significant tumour size divergence (Fig. 4i,j). Total TIL numbers were unchanged (Fig. 4k), but DFI increased the frequency of intratumoural CD8⁺ T cells and consequently lowered the CD4:CD8 ratio (Fig. 4l); SIINFEKL-loaded tetramers confirmed intratumoural enrichment of antigen-specific CD8⁺ cells (Fig. 4m and Supplementary Fig. 5i). In parallel, naive CD8⁺ cells expanded systemically (Supplementary Fig. 5j,k), suggesting that rejuvenated peripheral pools contribute to the TIL compartment.

scRNA–V(D)J profiling and CITE-seq using oligo-conjugated SIINFEKL–H-2K^b tetramers resolved eight canonical CD8⁺–CD4⁺ TIL states and delineated tumour antigen-specific T cells (Fig. 4n,o and Supplementary Fig. 5l–o). With DFI, both tumour-specific and bystander clonotypes showed lower expression of exhaustion-associated genes (*Havcr2*, *Gzmk*, *Lag3*, *Pdcd1* and *Tigit*) and moderately higher *Entpd1* (CD39) expression, consistent with increased tumour-antigen engagement or tissue residency rather than terminal dysfunction⁵¹ (Fig. 4p). The naive-like fraction among tetramer⁺ CD8⁺ TILs rose approximately 1.8-fold (Fig. 4q and Supplementary Fig. 5o), and clonal diversity (as assayed by Shannon index) increased in both tetramer⁺ and bystander repertoires (Fig. 4r and Supplementary Fig. 5p), indicating broader recruitment and intratumoural TCR repertoire breadth.

Together, DFI conditioning alone boosted endogenous antitumour control and synergized with PDL1 blockade to enhance therapeutic efficacy in aged hosts. Alongside recent work with mRNA-encoded cytokines⁶, these data highlight liver-encoded delivery of immune modulators as a strategy to mitigate age-related immune dysfunction and overcome components of immunotherapy resistance.

Immunological safety of DFI treatment

To define durability and safety of DFI, aged mice received DFI or control mRNA–LNPs for 28 days and were then observed off-treatment for 28 days (Supplementary Fig. 6a). Thymic output rose during dosing, as evidenced by elevated TCR excision circles, and returned to baseline thereafter (Supplementary Fig. 6b,c), consistent with the short half-life of mRNA and the rapid decline of thymocyte *Rag2* expression after a single dose (72 h or less; Fig. 2f and Extended Data Fig. 7c). STARmap of whole spleens showed that, despite waning thymic export, splenic T cell and dendritic cell numbers remained modestly elevated at day 28 post-cessation (Supplementary Fig. 6d–g), indicating partial persistence of peripheral remodelling. However, when vaccination occurred 4 weeks after stopping DFI, previously observed benefits such as higher vaccine-specific T cells, fewer exhaustion-like cells and favourable naive–effector–memory balance were no longer evident (Supplementary Fig. 6h–k). Thus, immune enhancement by our approach is largely confined to the dosing window and reverses upon withdrawal.

We next assessed whether DFI would exacerbate autoimmune disease. In NOD mice, which are prone to type 1 diabetes due to frequent generation of autoreactive TCRs, 4 weeks of DFI or control mRNA was followed by 6-month monitoring of glycaemia, glycosuria and

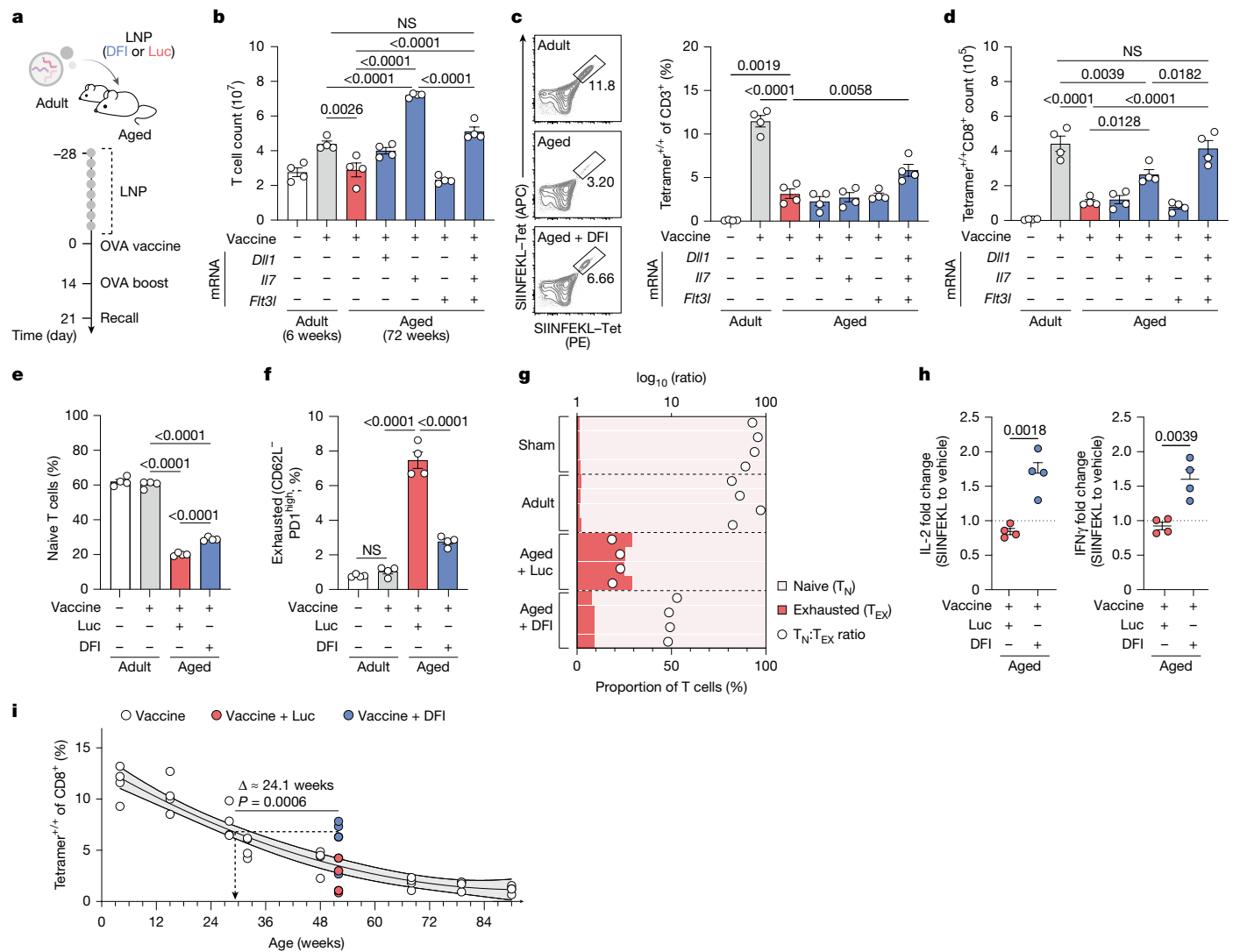


Fig. 3 | Hepatic DFI reconstitution enhances vaccine-induced T cell responses in aged hosts. **a**, Experimental design. Aged mice received DFI or control (Luc) mRNA-LNPs twice weekly for 28 days, then were immunized with adjuvanted OVA. **b**, Total splenic T cells (live CD45⁺CD3⁺) per spleen ($n = 4$ per group). Data are mean $\pm$ s.e.m.; one-way ANOVA with Tukey post-hoc test. **c**, Frequency of SIINFEKL-H-2K(b) tetramer⁺CD8⁺ T cells in the spleen ($n = 4$ per group). Data are mean $\pm$ s.e.m.; one-way ANOVA with Tukey post-hoc test. **d**, Absolute number of SIINFEKL-H-2K(b) tetramer⁺CD8⁺ T cells per spleen ($n = 4$ per group). Data are mean $\pm$ s.e.m.; one-way ANOVA with Tukey post-hoc test. **e**, Frequency of naive T cells (CD44⁺CD62L⁺) in the spleen ($n = 4$ per group). Data are mean $\pm$ s.e.m.; one-way ANOVA with Tukey post-hoc test. **f**, Frequency of exhausted-phenotype T cells (CD62L⁺PD1^{high}) in the spleen ($n = 4$ per group). Data are mean $\pm$ s.e.m.; one-way ANOVA with Tukey's post-hoc test. **g**, Proportions of naive (T_N) and exhausted (T_{EX}) T cells and the $T_N:T_{EX}$ ratio after repetitive OVA or sham (vehicle)

vaccination ($n = 4$ per treatment). **h**, Functional antigen recall: fold change in intracellular cytokines following 6-h SIINFEKL restimulation versus DMSO vehicle. IL-2 (left) and IFN γ (right) mean fluorescence intensity ($n = 4$ per group). Dotted lines indicate no change (fold change = 1.0); Data are mean $\pm$ s.e.m.; two-tailed unpaired t -tests. **i**, Age-response modelling. In a longitudinal cohort covering increasing ages, SIINFEKL-H-2K(b) tetramer frequencies were fit with a quadratic (second-order) polynomial regression with 95% confidence bands (d.f. = 29, $R^2 = 0.9116$). This standard curve was used to estimate an 'immunological vaccination response age' for numerically 52-week-old mice preconditioned with either DFI or Luc mRNA-LNPs before immunization ($N = 40$ total; $n = 4$ per age and treatment). The P value comparing the estimated vaccination response ages of the two treatment groups was calculated using a two-tailed unpaired t -test.

disease onset (Fig. 5a). In addition, we quantified frequencies of CD8⁺ T cells specific to the NRP-V7 mimotope, a known target in NOD mice³² (Extended Data Fig. 10a). DFI did not alter blood glucose levels, diabetes onset or autoreactive T cell frequencies versus control littermates (Fig. 5b,c), whereas NOD.Cg-Tg(TcrbNY8.3)1Pesa/DvsJ (NY8.3) NOD mice with a genetically encoded NRP-V7-restricted TCR uniformly developed diabetes within 8 weeks, providing a positive-control threshold. We did not observe any treatment-related adverse effects over the course of this 6-month study.

To test safety in a model with intact central tolerance, we used Act-mOVA mice, which constitutively express germline-encoded OVA³³ (Fig. 5d). This allowed us to use the same potent model antigen as in

our previous vaccination experiments; however, in this case, OVA was subject to central tolerance. Despite enhancing responses to exogenous OVA in aged wild-type mice, DFI neither induced OVA-specific CD4⁺ or CD8⁺ T cells after 4 weeks (Fig. 5e,f) nor broke humoral tolerance after adjuvanted OVA challenge in Act-mOVA mice, in contrast to wild-type controls (Fig. 5g).

Finally, in experimental autoimmune encephalitis, a polyclonal, antigen-driven central nervous system autoimmunity model, DFI was administered every 3 days following experimental autoimmune encephalitis induction by MOG₃₅₋₅₅ peptide immunization and thereafter until symptom onset. Aged mice showed delayed disease onset, consistent with impaired T cell priming and previous

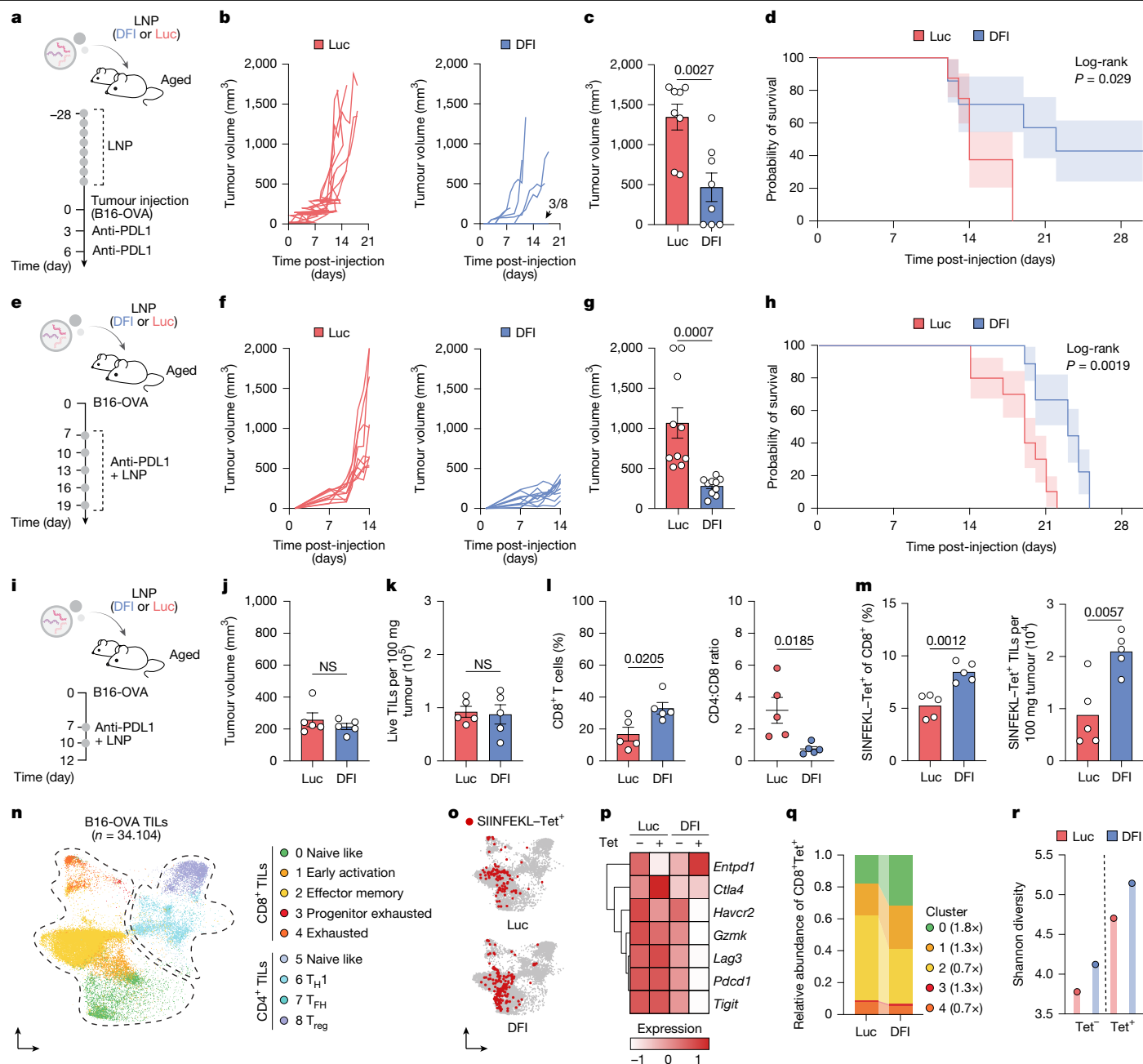


Fig. 4 | Hepatic DFI reconstitution enhances antitumour T cell responses in aged hosts. **a**, Experimental design. Aged mice received DFI or control (Luc) mRNA-LNPs for 28 days, followed by subcutaneous B16-OVA tumour challenge and two doses of anti-PDL1 checkpoint blockade. **b**, Tumour growth over time in the B16-OVA model ($n = 8$ per group). Three out of eight animals did not establish a measurable tumour (black arrow). **c**, Tumour size on day 12 (mean $\pm$ s.e.m.; two-tailed unpaired t -test). **d**, Kaplan-Meier survival analysis for the B16-OVA cohort ($n = 8$ per group; log-rank test). **e**, Design of adjuvant DFI or Luc LNP therapy combined with anti-PDL1 checkpoint blockade in established B16-OVA tumours. **f**, Tumour growth trajectories during combination therapy ($n = 10$ per group). **g**, Tumour size on day 14 (mean $\pm$ s.e.m.; two-tailed unpaired t -test). **h**, Kaplan-Meier survival curves for combination treatment ($n = 10$ per group; log-rank test). **i**, Schematic of a parallel adjuvant DFI + anti-PDL1 experiment terminated at day 12 for immune profiling with matched

tumour sizes. **j**, Tumour size at day 12 ($n = 5$ per group). **k**, Absolute counts of live TILs in explanted tumours ($n = 5$ per group). **l**, CD8⁺ T cell frequency (left) and CD4:CD8 ratio (right) among TILs ($n = 5$ per group). **m**, Relative (left) and absolute (right) numbers of tumour-specific SIINFEKL-H-2K(b)* TILs ($n = 5$ per group). For **j**–**m**, data are mean $\pm$ s.e.m.; statistical significance was determined by a two-tailed unpaired Student's t -test. **n**, Uniform manifold approximation and projection (UMAP) of scRNA-seq of CD45⁺CD3⁺ TILs ($n = 34,104$ cells). T_{FH}, follicular helper T; T_H, helper T; T_{reg}, regulatory T. **o**, CITE-seq feature plot showing CD8⁺SIINFEKL-H-2K(b)* (Tet⁺) TILs. **p**, Expression of canonical markers of tumour recognition and exhaustion in Luc-treated versus DFI-treated TILs. **q**, Relative abundance of transcriptional clusters among CD8⁺Tet⁺ clonotypes. All comparisons were by two-tailed unpaired t -tests (**o**–**q**). **r**, Shannon diversity of bystander (Tet⁻) and tumour-specific (Tet⁺) TCR clonotypes across treatments.

reports, but worse subsequent clinical deterioration^{54,55} (Extended Data Fig. 10b–d). In line with our earlier vaccination experiments, DFI increased peripheral, but not central nervous system-infiltrating MHC class III-A^b–MOG-specific CD4⁺ T cells after induction (Extended

Data Fig. 10e,f) and did not worsen clinical scores or spinal cord inflammation and demyelination (Extended Data Fig. 10d,g,h). In adults, DFI neither increased MOG-reactive T cells in the periphery nor aggravated disease (Extended Data Fig. 10c,e–h), indicating

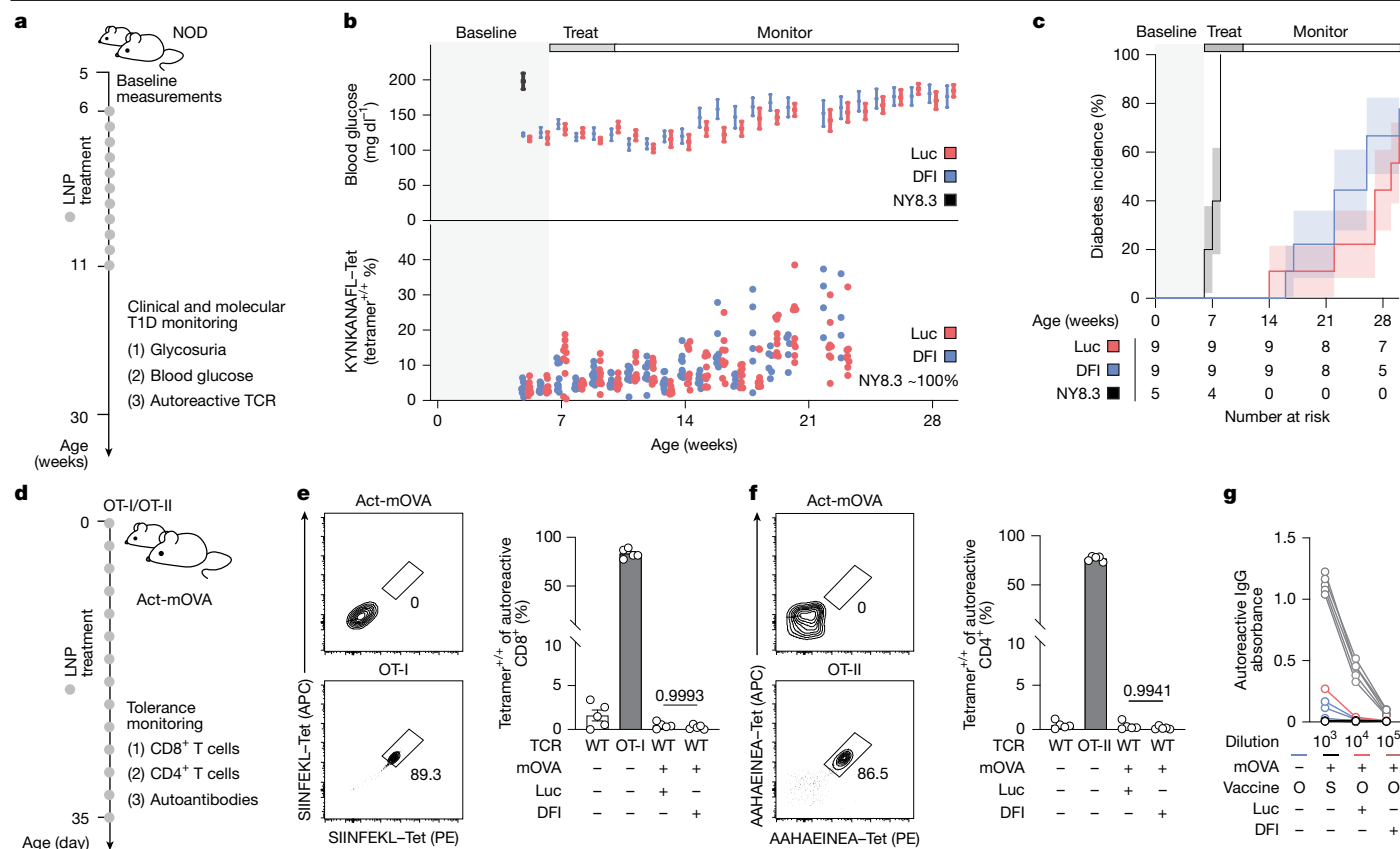


Fig. 5 | Immunological safety assessment of DFI. **a**, Experimental design to evaluate autoimmune risk in NOD mice receiving DFI or control (Luc) mRNA-LNPs. NY8.3 mice, carrying an autoreactive TCR specific for NRP-V7 (KYNKANAFI), served as positive controls. Type 1 diabetes (T1D) onset was defined by blood glucose > 200 mg dl⁻¹ in two consecutive measurements or by glucosuria. **b**, Longitudinal monitoring of blood glucose (top) and frequency of NRP-V7-specific TCRs in peripheral blood (bottom) of NOD mice treated as in panel **a**. NY8.3 ($n = 5$), NOD + Luc ($n = 9$) and NOD + DFI ($n = 9$). Data are mean $\pm$ s.e.m. **c**, Kaplan-Meier analysis of cumulative T1D incidence in NY8.3 ($n = 5$), NOD + Luc ($n = 9$) and NOD + DFI ($n = 9$) groups. Statistical significance was tested using log-rank (Mantel-Cox) tests. **d**, Experimental set-up to test antigen-specific T cell tolerance in Act-mOVA mice. OT-I (CD8⁺, OVA₂₅₇₋₂₆₄) and OT-II (CD4⁺, OVA₃₂₉₋₃₃₇) transgenic mice served as positive controls for

OVA-targeted responses. **e**, Frequencies of SIINFEKL-H2K(b) tetramer⁺ CD8⁺ T cells in peripheral blood of Act-mOVA mice treated with Luc or DFI mRNA for 28 days. $n = 5$ per group. Data are mean $\pm$ s.e.m.; one-way ANOVA with Tukey's post-hoc test; P values are shown for Act-mOVA comparisons. WT, wild type. **f**, Frequencies of AAHAIEINEA-I-A(b) tetramer⁺ CD4⁺ T cells in the peripheral blood of Act-mOVA mice treated as in panel **e**. $n = 5$ per group. Data are mean $\pm$ s.e.m.; one-way ANOVA with Tukey's post-hoc test; two-tailed P values are indicated for Act-mOVA comparisons. **g**, Serum anti-OVA IgG ELISA (OD₄₅₀₋₅₇₀) in Act-mOVA mice after Luc or DFI mRNA treatment followed by OVA-complete Freund's adjuvant prime and OVA-incomplete Freund's adjuvant boost (O) or sham vaccination (S). Wild-type ($n = 5$) vaccinated mice served as positive controls. Data are mean $\pm$ s.e.m.

correction of age-related deficits rather than indiscriminate immune amplification.

Together with the vaccination data, these results show that DFI transiently augments antigen-specific immunity in aged hosts while preserving self-tolerance across several models. The effects are reversible and temporally restricted, supporting liver-encoded systemic immune modulation as a novel strategy to improve immunity in older individuals.

Discussion

Here we have showed that immune function can be improved by repurposing the liver as a platform for ectopic production of immune factors. Our results show that delivery to the liver of mRNA encoding DLL1, IL-7 and FLT3-L in combination successfully enhanced immune function in aged mice. This approach provides a scalable alternative to more invasive methods of immune rejuvenation, such as thymic transplantation⁵⁶ or HSC manipulation¹², which are clinically challenging. In addition, our disease-agnostic strategy holds potential for synergizing with other immunostimulatory treatments, as we observed in the context of ICI treatment.

Although the biological roles of IL-7 and FLT3-L are well established, with both factors known to support thymopoiesis, dendritic cell expansion and peripheral T cell homeostasis^{20,21,35-37,41,42}, combining them with DLL1 and delivering them to hepatocytes enables the creation of a transient, rejuvenated immune milieu in aged mice, distinct from conventional approaches based on recombinant protein infusion, which lack spatial control and tend to require chronic dosing or come with high toxicity. By using mRNA, we mitigated the systemic inflammation and autoimmune sequelae typically associated with recombinant cytokine-based therapies. However, the transient nature of mRNA delivery necessitates repeated administrations to sustain therapeutic effects. The long-term consequences of continuous exposure to these factors, especially in aged individuals should be analysed through extensive long-term safety studies.

Ageing affects the immune system beyond T cell biology, encompassing epigenetic⁵⁷ and metabolic⁵⁸ remodelling in lymphocytes, dysfunction in myeloid subsets⁵⁹ and structural alterations in stromal networks⁶⁰. Future studies may therefore uncover additional strategies and factors to holistically target the hallmarks of immune ageing¹.

In summary, this study demonstrates that transiently repurposing the liver to express and secrete various therapeutic proteins could be

a generalizable approach to engineering physiological processes. By mimicking specific signalling niches within the liver or secreting proteins for systemic circulation to restore homeostatic signals throughout the body, this strategy has the potential to improve health outcomes and address a wide range of human diseases and conditions.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-025-09873-4>.

- Mittelbrunn, M. & Kroemer, G. Hallmarks of T cell aging. *Nat. Immunol.* **22**, 687–698 (2021).
- López-Otín, C., Pietrocola, F., Roiz-Valle, D., Galluzzi, L. & Kroemer, G. Meta-hallmarks of aging and cancer. *Cell Metab.* **35**, 12–35 (2023).
- Nikolich-Zugich, J. The twilight of immunity: emerging concepts in aging of the immune system. *Nat. Immunol.* **19**, 10–19 (2018).
- Ventevogel, M. S. & Sempowski, G. D. Thymic rejuvenation and aging. *Curr. Opin. Immunol.* **25**, 516–522 (2013).
- Chaudhry, M. S., Velardi, E., Dudakov, J. A. & van den Brink, M. R. M. Thymus: the next (re) generation. *Immunol. Rev.* **271**, 56–71 (2016).
- Beck, J. D. et al. Long-lasting mRNA-encoded interleukin-2 restores CD8⁺ T cell neoantigen immunity in MHC class I-deficient cancers. *Cancer Cell* **42**, 568–582.e11 (2024).
- Quijano-Rubio, A. et al. A split, conditionally active mimetic of IL-2 reduces the toxicity of systemic cytokine therapy. *Nat. Biotechnol.* **41**, 532–540 (2023).
- Hince, M. et al. The role of sex steroids and gonadectomy in the control of thymic involution. *Cell. Immunol.* **252**, 122–138 (2008).
- Cosway, E. J. et al. The alarmin IL33 orchestrates type 2 immune-mediated control of thymus regeneration. *Nat. Commun.* **14**, 7201 (2023).
- Sharma, H. & Moroni, L. Recent advancements in regenerative approaches for thymus rejuvenation. *Adv. Sci.* **8**, 2100543 (2021).
- Yankova, T., Dubiley, T., Shytikov, D. & Pishel, I. Three month heterochronic parabiosis has a deleterious effect on the lifespan of young animals, without a positive effect for old animals. *Rejuvenation Res.* **25**, 191–199 (2022).
- Ross, J. B. et al. Depleting myeloid-biased hematopoietic stem cells rejuvenates aged immunity. *Nature* **628**, 162–170 (2024).
- Savino, W., Mendes-da-Cruz, D. A., Lepletier, A. & Dardenne, M. Hormonal control of T-cell development in health and disease. *Nat. Rev. Endocrinol.* **12**, 77–89 (2016).
- Kumar, B. V., Connors, T. J. & Farber, D. L. Human T cell development, localization, and function throughout life. *Immunity* **48**, 202–213 (2018).
- Farber, D. L., Yudanin, N. A. & Restifo, N. P. Human memory T cells: generation, compartmentalization and homeostasis. *Nat. Rev. Immunol.* **14**, 24–35 (2014).
- Liu, S. et al. Tissue architecture dynamics underlying immune development and decline in the thymus. Preprint at *bioRxiv* <https://doi.org/10.1101/2025.06.28.662120> (2025).
- Mogilenko, D. A., Shchukina, I. & Artyomov, M. N. Immune ageing at single-cell resolution. *Nat. Rev. Immunol.* **22**, 484–498 (2022).
- Mogilenko, D. A. Comprehensive profiling of an aging immune system reveals clonal GZMK⁺ CD8⁺ T cells as conserved hallmark of inflammaging. *Immunity* **54**, 99–115 (2021).
- Rochman, Y., Spolski, R. & Leonard, W. J. New insights into the regulation of T cells by yc family cytokines. *Nat. Rev. Immunol.* **9**, 480–490 (2009).
- Kenins, L., Gill, J. W., Boyd, R. L., Holländer, G. A. & Wodnar-Filipowicz, A. Intrathymic expression of Flt3 ligand enhances thymic recovery after irradiation. *J. Exp. Med.* **205**, 523–531 (2008).
- Williams, K. M. et al. FLT3 ligand regulates thymic precursor cells and hematopoietic stem cells through interactions with CXCR4 and the marrow niche. *Exp. Hematol.* **52**, 40–49 (2017).
- Hellström, M. et al. Dll4 signalling through Notch1 regulates formation of tip cells during angiogenesis. *Nature* **445**, 776–780 (2007).
- Morell, C. M. & Strazzabosco, M. Notch signaling and new therapeutic options in liver disease. *J. Hepatol.* **60**, 885–890 (2014).
- Warren, A. et al. T lymphocytes interact with hepatocytes through fenestrations in murine liver sinusoidal endothelial cells. *Hepatology* **44**, 1182–1190 (2006).
- Kawashima, K., Andreato, F., Beccaria, C. G. & Iannaccone, M. Priming and maintenance of adaptive immunity in the liver. *Annu. Rev. Immunol.* <https://doi.org/10.1146/annurev-immunol-090122-041354> (2024).
- Ward, W. & Richardson, A. Effect of age on liver protein synthesis and degradation. *Hepatology* **14**, 935–948 (1991).
- Sahin, U., Karikó, K. & Türeci, Ö. mRNA-based therapeutics-developing a new class of drugs. *Nat. Rev. Drug Discov.* **13**, 759–780 (2014).
- Karikó, K. et al. Incorporation of pseudouridine into mRNA yields superior nonimmunogenic vector with increased translational capacity and biological stability. *Mol. Ther.* **16**, 1833–1840 (2008).
- Chen, H. et al. Branched chemically modified poly(A) tails enhance the translation capacity of mRNA. *Nat. Biotechnol.* **43**, 194–203 (2025).
- Zeng, H. et al. Spatially resolved single-cell translationalomics at molecular resolution. *Science* **380**, eadd3067 (2023).
- Scavuzzo, M. A. et al. Pancreatic cell fate determination relies on Notch ligand trafficking by NFIA. *Cell Rep.* **25**, 3811–3827.e7 (2018).
- Roberts, R. et al. Heparan sulphate bound growth factors: a mechanism for stromal cell mediated haemopoiesis. *Nature* **332**, 376–378 (1988).
- Ariel, A. et al. Induction of T cell adhesion to extracellular matrix or endothelial cell ligands by soluble or matrix-bound interleukin-7. *Eur. J. Immunol.* **27**, 2562–2570 (1997).
- Stadler, C. R. et al. Elimination of large tumors in mice by mRNA-encoded bispecific antibodies. *Nat. Med.* **23**, 815–817 (2017).
- Schluns, K. S., Kieper, W. C., Jameson, S. C. & Lefrançois, L. Interleukin-7 mediates the homeostasis of naïve and memory CD8 T cells in vivo. *Nat. Immunol.* **1**, 426–432 (2000).
- Fry, T. J. & Mackall, C. L. The many faces of IL-7: from lymphopoiesis to peripheral T cell maintenance. *J. Immunol.* **174**, 6571–6576 (2005).
- Mackall, C. L., Fry, T. J. & Gress, R. E. Harnessing the biology of IL-7 for therapeutic application. *Nat. Rev. Immunol.* **11**, 330–342 (2011).
- Rossi, D. J. et al. Cell intrinsic alterations underlie hematopoietic stem cell aging. *Proc. Natl Acad. Sci. USA* **102**, 9194–9199 (2005).
- Pang, W. W. et al. Human bone marrow hematopoietic stem cells are increased in frequency and myeloid-biased with age. *Proc. Natl Acad. Sci. USA* **108**, 20012–20017 (2011).
- Zlotoff, D. A. et al. CCR7 and CCR9 together recruit hematopoietic progenitors to the adult thymus. *Blood* **115**, 1897–1905 (2010).
- Karsunky, H., Merad, M., Cozzio, A., Weissman, I. L. & Manz, M. G. Flt3 ligand regulates dendritic cell development from Flt3⁺ lymphoid and myeloid-committed progenitors to Flt3⁺ dendritic cells in vivo. *J. Exp. Med.* **198**, 305–313 (2003).
- Waskow, C. et al. The receptor tyrosine kinase Flt3 is required for dendritic cell development in peripheral lymphoid tissues. *Nat. Immunol.* **9**, 676–683 (2008).
- Shi, H. et al. Spatial atlas of the mouse central nervous system at molecular resolution. *Nature* **622**, 552–561 (2023).
- Hao, Y., O'Neill, P., Naradikian, M. S., Scholz, J. L. & Cancro, M. P. A B-cell subset uniquely responsive to innate stimuli accumulates in aged mice. *Blood* **118**, 1294–1304 (2011).
- Rubtsova, K., Rubtsov, A. V., Cancro, M. P. & Marrack, P. Age-associated B cells: a T-bet-dependent effector with roles in protective and pathogenic immunity. *J. Immunol.* **195**, 1933–1937 (2015).
- Collier, D. A. et al. Age-related immune response heterogeneity to SARS-CoV-2 vaccine BNT162b2. *Nature* **596**, 417–422 (2021).
- Elyahu, Y. et al. Aging promotes reorganization of the CD4 T cell landscape toward extreme regulatory and effector phenotypes. *Sci. Adv.* **5**, eaaw8330 (2019).
- Zhivaki, D. et al. Correction of age-associated defects in dendritic cells enables CD4⁺ T cells to eradicate tumors. *Cell* **187**, 3888–3903.e18 (2024).
- Tsakamoto, H. et al. Aging-associated and CD4 T-cell-dependent ectopic CXCL13 activation predisposes to anti-PD-1 therapy-induced adverse events. *Proc. Natl Acad. Sci. USA* **119**, e2205378119 (2022).
- Georgiev, P. et al. Age-associated contraction of tumor-specific T cells impairs antitumor immunity. *Cancer Immunol. Res.* **12**, 1525–1541 (2024).
- Duhen, T. et al. Co-expression of CD39 and CD103 identifies tumor-reactive CD8 T cells in human solid tumors. *Nat. Commun.* **9**, 2724 (2018).
- Trudeau, J. H. D. et al. Prediction of spontaneous autoimmune diabetes in NOD mice by quantification of autoreactive T cells in peripheral blood. *J. Clin. Invest.* **111**, 217–223 (2003).
- Ehst, B. D., Ingulli, E. & Jenkins, M. K. Development of a novel transgenic mouse for the study of interactions between CD4 and CD8 T cells during graft rejection. *Am. J. Transplant.* **3**, 1355–1362 (2003).
- Aharoni, R. et al. Age dependent course of EAE in Aire^{-/-} mice. *J. Neuroimmunol.* **262**, 27–34 (2013).
- Atkinson, J. R. et al. Biological aging of CNS-resident cells alters the clinical course and immunopathology of autoimmune demyelinating disease. *JCI Insight* <https://doi.org/10.1172/jci.insight.158153> (2022).
- Nobori, S. et al. Thymic rejuvenation and the induction of tolerance by adult thymic grafts. *Proc. Natl Acad. Sci. USA* **103**, 19081–19086 (2006).
- Desdín-Micó, G. et al. T cells with dysfunctional mitochondria induce multimorbidity and premature senescence. *Science* **368**, 1371–1376 (2020).
- Han, S., Georgiev, P., Ringel, A. E., Sharpe, A. H. & Haigis, M. C. Age-associated remodeling of T cell immunity and metabolism. *Cell Metab.* **35**, 36–55 (2023).
- Park, M. D. et al. Hematopoietic aging promotes cancer by fueling IL-1α-driven emergency myelopoiesis. *Science* **386**, eadn0327 (2024).
- Sladitschek-Martens, H. L. et al. YAP/TAZ activity in stromal cells prevents ageing by controlling cGAS-STING. *Nature* **607**, 790–798 (2022).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

© The Author(s) 2025